Zain Usmani

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Education

University of Windsor 2024 - Present
BASc in Electrical and Computer Engineering (Minor in CS) GPA: 3.7/4.0

Experience

Embedded Software Lead Dec 2024 – Present
Formula Electric - Lancer Motorsport

· Architected a real-time vehicle telemetry pipeline logging 11 dynamics channels at 50 Hz
· Built Python regression tests querying SQLite databases to validate telemetry data integrity
· Automated SQLite telemetry parsing with Python Pandas to deliver rapid performance reports
· Managed a continuous integration pipeline gating every Git merge to main on hardware unit tests
· Designed software testing frameworks to isolate and validate custom C drivers before deployment
· Debugged memory leaks in a Linux QT dashboard by enforcing strict pointer lifecycle management

Software Engineering Co-op Apr 2026 – Aug 2026
O-Hive

· Engineered an end-to-end data pipeline converting 2D video into metric-scale 3D point clouds
· Migrated a cloud rendering pipeline to AnySplat to sustain 15 fps under a 14 GB VRAM ceiling
· Containerized an Ubuntu 22.04 workspace via Docker to standardize dependencies for an FPGA board
· Automated deployments via Bash scripts to execute cross-compiled binaries on embedded Linux
· Wrote a C++ memory fork splitting NV12 frame buffers across CPU and FPGA compute paths
· Debugged hardware integration bottlenecks by parsing Linux system logs across the data pipeline
· Managed Git branching and version control across cross-functional hardware and software teams

ML and Data Research Intern May 2025 – Dec 2025
MicroNano Mechatronics Lab

· Streamed live 50 Hz IMU data in C++ firmware with error handling for packet loss and latency
· Engineered automated Python data sanitization scripts to clean sensor readings for training
· Analyzed 16000 samples with Pandas and NumPy, training XGBoost models to a 78.9 F1 score

Projects

LLM Automated Generation Pipeline | *Python, Anthropic API, LaTeX, PowerShell* Oct 2026

· Engineered a headless Python and PowerShell automation pipeline integrating the Anthropic API
· Enforced strict JSON metadata extraction via prompt guardrails to block malformed artifacts
· Built a Python regex routing script to sanitize company names and archive compiled PDF output

RouteScout Rover (McHacks 13) | *ROS2, C++, ESP32, RPi 5* Jan 2026

· Containerized a ROS2 environment managing sensor data streams into path planning algorithms
· Fused wheel encoder and IMU data at 50 Hz on an ESP32 and relayed odometry to an RPi 5 node

Autonomous Fire Extinguisher | *Linux, RPi 4B, RPi Cam, YOLOv4* Feb 2026

· Configured Linux on an RPi 4B to interface a camera and execute continuous inference loops
· Deployed a YOLOv4 hazard detection pipeline on Linux dispatching real-time triggers to an MCU

Skills

Languages and Tools: C++, C, Python, Shell Scripting, Linux (Ubuntu), Git, CMake
Data and Infrastructure: Docker, SQLite, Pandas, NumPy, Regression Testing, CI Pipelines
Embedded and ML: AMD Kria KV260, ESP32, Raspberry Pi, PyTorch, OpenCV, ONNX, XGBoost
Concepts: OOP, Continuous Integration, Real-Time Telemetry, Data Integrity, Memory Management